Supporting Information

Prevalence-adjusted and temporally ordered associations among respiratory pathogens in a multicentre hospital surveillance network in China: a retrospective observational study

Authors

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Table S1: Pathogen positivity by age group

Pathogen	<5 years	5–17 years	18-60 years	>60 years	P value
Virus	**28,701/231,436 (12·40%)**	**64,724/484,630 (13·36%)**	**51,030/428,803 (11·90%)**	**35,690/374,203 (9·54%)**	**p<0·0001**
CMV	19/93 (20·43%)	48/136 (35·29%)	127/596 (21·31%)	211/1,233 (17·11%)	p<0·0001
SARS-CoV-2	1,970/23,577 (8·36%)	4,802/51,523 (9·32%)	14,893/90,745 (16·41%)	11,756/78,022 (15·07%)	p<0·0001
EBV	149/465 (32·04%)	547/1,460 (37·47%)	1,005/4,215 (23·84%)	1,456/4,198 (34·68%)	p<0·0001
HAdV	1,333/29,028 (4·59%)	2,884/65,063 (4·43%)	1,367/59,482 (2·30%)	1,156/53,266 (2·17%)	p<0·0001
HBoV	238/12,752 (1·87%)	153/20,554 (0·74%)	11/5,483 (0·20%)	11/10,157 (0·11%)	p<0·0001
HCoV	272/7,213 (3·77%)	524/12,417 (4·22%)	761/4,600 (16·54%)	734/6,016 (12·20%)	p<0·0001
HMPV	469/5,435 (8·63%)	441/8,532 (5·17%)	177/4,620 (3·83%)	309/9,217 (3·35%)	p<0·0001
HPIV	1,589/20,362 (7·80%)	1,744/37,885 (4·60%)	975/22,415 (4·35%)	1,163/30,497 (3·81%)	p<0·0001
HRV	3,741/14,185 (26·37%)	4,699/21,788 (21·57%)	693/7,988 (8·68%)	941/14,309 (6·58%)	p<0·0001
HSV	3/80 (3·75%)	14/152 (9·21%)	41/257 (15·95%)	115/336 (34·23%)	p<0·0001
IFV	15,035/85,349 (17·62%)	45,816/203,120 (22·56%)	29,608/182,074 (16·26%)	15,974/115,879 (13·79%)	p<0·0001
RSV	3,883/32,897 (11·80%)	3,052/62,000 (4·92%)	1,372/46,328 (2·96%)	1,864/51,073 (3·65%)	p<0·0001
Atypical bacteria	**15,018/88,941 (16·89%)**	**41,041/159,915 (25·66%)**	**10,550/127,566 (8·27%)**	**4,098/95,726 (4·28%)**	**p<0·0001**
MP	14,869/75,471 (19·70%)	40,195/129,874 (30·95%)	9,877/95,626 (10·33%)	3,670/73,054 (5·02%)	p<0·0001
CP	149/13,470 (1·11%)	846/30,041 (2·82%)	673/31,940 (2·11%)	428/22,672 (1·89%)	p<0·0001
Bacteria	**3,185/17,742 (17·95%)**	**4,344/28,673 (15·15%)**	**7,463/30,172 (24·73%)**	**14,835/55,965 (26·51%)**	**p<0·0001**
Bordetella pertussis	375/7,475 (5·02%)	810/9,305 (8·70%)	71/1,720 (4·13%)	23/2,251 (1·02%)	p<0·0001
Haemophilus influenzae	2,159/4,537 (47·59%)	2,023/6,173 (32·77%)	2,021/5,054 (39·99%)	3,908/10,865 (35·97%)	p<0·0001
Klebsiella pneumoniae	47/91 (51·65%)	83/157 (52·87%)	401/824 (48·67%)	1,198/1,873 (63·96%)	p<0·0001
Legionella	1/1,320 (0·08%)	25/3,330 (0·75%)	151/3,578 (4·22%)	130/5,416 (2·40%)	p<0·0001
Moraxella catarrhalis	82/430 (19·07%)	85/850 (10·00%)	56/901 (6·22%)	104/1,215 (8·56%)	p<0·0001
MTB	203/791 (25·66%)	539/2,600 (20·73%)	2,920/10,158 (28·75%)	3,968/15,625 (25·40%)	p<0·0001
Pseudomonas aeruginosa	24/559 (4·29%)	57/689 (8·27%)	262/956 (27·41%)	918/2,611 (35·16%)	p<0·0001
Staphylococcus aureus	73/969 (7·53%)	177/2,230 (7·94%)	441/3,115 (14·16%)	1,260/7,659 (16·45%)	p<0·0001
Stenotrophomonas maltophilia	4/143 (2·80%)	43/422 (10·19%)	264/1,007 (26·22%)	1,041/2,302 (45·22%)	p<0·0001
Streptococcus pneumoniae	200/1,275 (15·69%)	428/2,482 (17·24%)	199/1,449 (13·73%)	231/2,657 (8·69%)	p<0·0001
Acinetobacter baumannii	17/152 (11·18%)	74/435 (17·01%)	677/1,410 (48·01%)	2,054/3,491 (58·84%)	p<0·0001
Fungi	**960/1,057 (90·82%)**	**1,667/1,866 (89·34%)**	**368/2,502 (14·71%)**	**898/6,195 (14·50%)**	**p<0·0001**
Aspergillus	0/7 (0·00%)	0/24 (0·00%)	11/271 (4·06%)	75/1,086 (6·91%)	NS
Candida	954/989 (96·46%)	1,665/1,712 (97·25%)	286/565 (50·62%)	714/1,225 (58·29%)	p<0·0001
Cryptococcus	3/44 (6·82%)	2/111 (1·80%)	42/1,377 (3·05%)	42/2,915 (1·44%)	p=0·00053
Pneumocystis	3/17 (17·65%)	0/19 (0·00%)	29/289 (10·03%)	67/969 (6·91%)	NS

Data are positive tests/total tests (%). Denominators vary by pathogen and age group because testing was clinically directed. Table entries report observed rates; only the heatmap in Supplementary figure 2B displays pathogen–stratum combinations with fewer than 100 tested encounters as 0%. CMV=cytomegalovirus; SARS-CoV-2=severe acute respiratory syndrome coronavirus 2; EBV=Epstein–Barr virus; HAdV=human adenovirus; HBoV=human bocavirus; HCoV=seasonal human coronavirus; HMPV=human metapneumovirus; HPIV=human parainfluenza virus; HRV=human rhinovirus; HSV=herpes simplex virus; IFV=influenza virus; RSV=respiratory syncytial virus; MP=Mycoplasma pneumoniae; CP=Chlamydia pneumoniae; MTB=Mycobacterium tuberculosis; NS=not significant.

Table S2: Pathogen positivity by sex

Pathogen	Male	Female	P value
Virus	**120,203/1,018,537 (11·80%)**	**107,894/896,169 (12·04%)**	**p<0·0001**
CMV	285/1,358 (20·99%)	214/908 (23·57%)	NS
SARS-CoV-2	19,923/155,074 (12·85%)	21,345/141,243 (15·11%)	p<0·0001
EBV	2,210/7,478 (29·55%)	1,598/6,718 (23·79%)	p<0·0001
HAdV	4,674/138,015 (3·39%)	3,998/126,539 (3·16%)	p=0·0011
HBoV	569/37,104 (1·53%)	360/25,604 (1·41%)	NS
HCoV	1,363/23,030 (5·92%)	1,278/16,213 (7·88%)	p<0·0001
HMPV	932/19,134 (4·87%)	739/12,822 (5·76%)	p=0·00049
HPIV	4,184/83,625 (5·00%)	3,334/67,002 (4·98%)	NS
HRV	8,342/43,759 (19·06%)	5,438/31,290 (17·38%)	p<0·0001
HSV	127/547 (23·22%)	58/407 (14·25%)	p=0·00072
IFV	68,462/370,971 (18·45%)	62,511/346,528 (18·04%)	p<0·0001
RSV	9,132/138,442 (6·60%)	7,021/120,895 (5·81%)	p<0·0001
Atypical bacteria	**43,407/298,441 (14·54%)**	**43,526/271,054 (16·06%)**	**p<0·0001**
MP	42,370/243,175 (17·42%)	42,380/219,193 (19·33%)	p<0·0001
CP	1,037/55,266 (1·88%)	1,146/51,861 (2·21%)	p=0·00012
Bacteria	**23,947/101,385 (23·62%)**	**14,819/71,747 (20·65%)**	**p<0·0001**
Bordetella pertussis	875/15,776 (5·55%)	667/12,068 (5·53%)	NS
Haemophilus influenzae	9,651/22,432 (43·02%)	5,924/15,735 (37·65%)	p<0·0001
Klebsiella pneumoniae	1,502/2,557 (58·74%)	713/1,579 (45·16%)	p<0·0001
Legionella	181/9,381 (1·93%)	140/7,473 (1·87%)	NS
Moraxella catarrhalis	209/2,324 (8·99%)	179/1,639 (10·92%)	NS
MTB	5,020/22,063 (22·75%)	3,753/14,641 (25·63%)	p<0·0001
Pseudomonas aeruginosa	1,006/3,752 (26·81%)	505/2,468 (20·46%)	p<0·0001
Staphylococcus aureus	1,397/10,203 (13·69%)	988/8,013 (12·33%)	p=0·0073
Stenotrophomonas maltophilia	1,079/2,944 (36·65%)	444/1,642 (27·04%)	p<0·0001
Streptococcus pneumoniae	871/5,865 (14·85%)	546/4,248 (12·85%)	p=0·0047
Acinetobacter baumannii	2,156/4,088 (52·74%)	960/2,241 (42·84%)	p<0·0001
Fungi	**3,496/8,836 (39·57%)**	**2,609/6,437 (40·53%)**	**NS**
Aspergillus	61/968 (6·30%)	43/721 (5·96%)	NS
Candida	3,280/3,961 (82·81%)	2,499/3,058 (81·72%)	NS
Cryptococcus	66/2,987 (2·21%)	37/2,000 (1·85%)	NS
Pneumocystis	89/920 (9·67%)	30/658 (4·56%)	p=0·00022

Data are positive tests/total tests (%). Records with unknown sex are not shown. Denominators vary by pathogen because testing was clinically directed. CMV=cytomegalovirus; SARS-CoV-2=severe acute respiratory syndrome coronavirus 2; EBV=Epstein–Barr virus; HAdV=human adenovirus; HBoV=human bocavirus; HCoV=seasonal human coronavirus; HMPV=human metapneumovirus; HPIV=human parainfluenza virus; HRV=human rhinovirus; HSV=herpes simplex virus; IFV=influenza virus; RSV=respiratory syncytial virus; MP=Mycoplasma pneumoniae; CP=Chlamydia pneumoniae; MTB=Mycobacterium tuberculosis; NS=not significant.

Table S3: Pathogen positivity by season

Pathogen	Spring	Summer	Autumn	Winter	P value
Virus	**48,056/454,946 (10·56%)**	**36,504/358,116 (10·19%)**	**48,419/409,493 (11·82%)**	**88,197/604,154 (14·60%)**	**p<0·0001**
CMV	118/655 (18·02%)	190/723 (26·28%)	77/348 (22·13%)	113/533 (21·20%)	p=0·0030
SARS-CoV-2	12,247/76,945 (15·92%)	12,253/64,574 (18·98%)	5,830/57,383 (10·16%)	8,463/85,473 (9·90%)	p<0·0001
EBV	1,074/3,660 (29·34%)	881/3,706 (23·77%)	616/2,186 (28·18%)	1,086/3,598 (30·18%)	p<0·0001
HAdV	2,018/60,821 (3·32%)	1,878/49,644 (3·78%)	2,087/60,591 (3·44%)	2,542/80,501 (3·16%)	p<0·0001
HBoV	76/16,320 (0·47%)	353/14,357 (2·46%)	360/14,472 (2·49%)	136/16,349 (0·83%)	p<0·0001
HCoV	739/10,610 (6·97%)	661/9,639 (6·86%)	425/8,661 (4·91%)	690/9,201 (7·50%)	p<0·0001
HMPV	575/8,638 (6·66%)	53/6,254 (0·85%)	391/6,877 (5·69%)	625/9,065 (6·89%)	p<0·0001
HPIV	2,278/36,075 (6·31%)	1,879/29,026 (6·47%)	1,612/34,217 (4·71%)	1,530/40,143 (3·81%)	p<0·0001
HRV	3,827/19,393 (19·73%)	2,890/19,758 (14·63%)	4,359/16,352 (26·66%)	2,563/18,681 (13·72%)	p<0·0001
HSV	48/201 (23·88%)	45/211 (21·33%)	51/271 (18·82%)	40/268 (14·93%)	NS
IFV	20,561/158,483 (12·97%)	12,767/115,718 (11·03%)	29,129/147,942 (19·69%)	65,351/264,139 (24·74%)	p<0·0001
RSV	4,495/63,145 (7·12%)	2,654/44,506 (5·96%)	3,482/60,193 (5·78%)	5,058/76,203 (6·64%)	p<0·0001
Atypical bacteria	**15,942/126,321 (12·62%)**	**20,038/120,284 (16·66%)**	**24,015/142,882 (16·81%)**	**24,837/158,743 (15·65%)**	**p<0·0001**
MP	15,535/105,310 (14·75%)	19,642/97,663 (20·11%)	23,485/113,277 (20·73%)	24,102/129,908 (18·55%)	p<0·0001
CP	407/21,011 (1·94%)	396/22,621 (1·75%)	530/29,605 (1·79%)	735/28,835 (2·55%)	p<0·0001
Bacteria	**9,288/47,360 (19·61%)**	**6,361/35,450 (17·94%)**	**7,197/33,874 (21·25%)**	**9,680/43,885 (22·06%)**	**p<0·0001**
Bordetella pertussis	454/8,189 (5·54%)	727/7,429 (9·79%)	182/6,327 (2·88%)	174/5,742 (3·03%)	p<0·0001
Haemophilus influenzae	3,129/10,206 (30·66%)	1,187/5,544 (21·41%)	2,658/7,381 (36·01%)	2,725/8,258 (33·00%)	p<0·0001
Klebsiella pneumoniae	593/1,188 (49·92%)	559/1,173 (47·66%)	556/912 (60·96%)	480/831 (57·76%)	p<0·0001
Legionella	79/3,517 (2·25%)	42/2,047 (2·05%)	89/3,512 (2·53%)	100/4,387 (2·28%)	NS
Moraxella catarrhalis	128/1,332 (9·61%)	99/855 (11·58%)	75/666 (11·26%)	84/1,085 (7·74%)	p=0·019
MTB	1,956/9,750 (20·06%)	1,274/8,273 (15·40%)	1,671/7,420 (22·52%)	3,713/10,901 (34·06%)	p<0·0001
Pseudomonas aeruginosa	413/1,927 (21·43%)	351/1,118 (31·40%)	379/927 (40·88%)	348/2,205 (15·78%)	p<0·0001
Staphylococcus aureus	666/4,867 (13·68%)	595/4,048 (14·70%)	460/2,891 (15·91%)	568/5,308 (10·70%)	p<0·0001
Stenotrophomonas maltophilia	377/1,272 (29·64%)	496/1,293 (38·36%)	323/796 (40·58%)	322/919 (35·04%)	p<0·0001
Streptococcus pneumoniae	603/3,303 (18·26%)	218/2,006 (10·87%)	231/1,978 (11·68%)	357/2,780 (12·84%)	p<0·0001
Acinetobacter baumannii	890/1,809 (49·20%)	813/1,664 (48·86%)	573/1,064 (53·85%)	809/1,469 (55·07%)	p=0·00038
Fungi	**2,356/5,260 (44·79%)**	**1,061/3,770 (28·14%)**	**1,798/3,016 (59·62%)**	**811/2,859 (28·37%)**	**p<0·0001**
Aspergillus	29/542 (5·35%)	26/544 (4·78%)	14/171 (8·19%)	24/305 (7·87%)	NS
Candida	2,249/2,636 (85·32%)	989/1,238 (79·89%)	1,740/1,881 (92·50%)	743/1,199 (61·97%)	p<0·0001
Cryptococcus	42/1,557 (2·70%)	20/1,470 (1·36%)	19/731 (2·60%)	16/1,094 (1·46%)	p=0·020
Pneumocystis	36/525 (6·86%)	26/518 (5·02%)	25/233 (10·73%)	28/261 (10·73%)	p=0·0063

Data are positive tests/total tests (%). Denominators vary by pathogen and season because testing was clinically directed. CMV=cytomegalovirus; SARS-CoV-2=severe acute respiratory syndrome coronavirus 2; EBV=Epstein–Barr virus; HAdV=human adenovirus; HBoV=human bocavirus; HCoV=seasonal human coronavirus; HMPV=human metapneumovirus; HPIV=human parainfluenza virus; HRV=human rhinovirus; HSV=herpes simplex virus; IFV=influenza virus; RSV=respiratory syncytial virus; MP=Mycoplasma pneumoniae; CP=Chlamydia pneumoniae; MTB=Mycobacterium tuberculosis; NS=not significant.

Table S4: Pathogen positivity by geographic region

Pathogen	North	South	P value
Virus	**54,297/432,421 (12·56%)**	**174,005/1,482,945 (11·73%)**	**p<0·0001**
CMV	234/1,059 (22·10%)	266/1,208 (22·02%)	NS
SARS-CoV-2	8,881/77,229 (11·50%)	32,503/219,332 (14·82%)	p<0·0001
EBV	2,240/4,183 (53·55%)	1,570/10,015 (15·68%)	p<0·0001
HAdV	2,293/52,598 (4·36%)	6,381/211,981 (3·01%)	p<0·0001
HBoV	214/20,682 (1·03%)	715/42,027 (1·70%)	p<0·0001
HCoV	145/3,875 (3·74%)	2,497/35,371 (7·06%)	p<0·0001
HMPV	621/12,922 (4·81%)	1,050/19,034 (5·52%)	p=0·0055
HPIV	3,229/47,495 (6·80%)	4,289/103,137 (4·16%)	p<0·0001
HRV	2,578/14,401 (17·90%)	11,202/60,649 (18·47%)	NS
HSV	19/48 (39·58%)	166/906 (18·32%)	p=0·00057
IFV	29,068/144,759 (20·08%)	101,985/573,080 (17·80%)	p<0·0001
RSV	4,775/53,170 (8·98%)	11,381/206,205 (5·52%)	p<0·0001
Atypical bacteria	**19,622/102,183 (19·20%)**	**67,323/467,355 (14·41%)**	**p<0·0001**
MP	19,063/84,349 (22·60%)	65,698/378,057 (17·38%)	p<0·0001
CP	559/17,834 (3·13%)	1,625/89,298 (1·82%)	p<0·0001
Bacteria	**7,407/53,980 (13·72%)**	**31,359/119,167 (26·32%)**	**p<0·0001**
Bordetella pertussis	91/1,411 (6·45%)	1,451/26,436 (5·49%)	NS
Haemophilus influenzae	1,139/6,630 (17·18%)	14,436/31,541 (45·77%)	p<0·0001
Klebsiella pneumoniae	658/2,002 (32·87%)	1,557/2,134 (72·96%)	p<0·0001
Legionella	289/11,778 (2·45%)	32/5,077 (0·63%)	p<0·0001
Moraxella catarrhalis	40/760 (5·26%)	348/3,203 (10·86%)	p<0·0001
MTB	1,339/12,819 (10·45%)	7,434/23,886 (31·12%)	p<0·0001
Pseudomonas aeruginosa	296/3,898 (7·59%)	1,215/2,322 (52·33%)	p<0·0001
Staphylococcus aureus	538/6,389 (8·42%)	1,847/11,831 (15·61%)	p<0·001
Stenotrophomonas maltophilia	799/1,551 (51·52%)	724/3,035 (23·86%)	p<0·0001
Streptococcus pneumoniae	438/4,209 (10·41%)	979/5,906 (16·58%)	p<0·0001
Acinetobacter baumannii	1,780/2,533 (70·27%)	1,336/3,796 (35·19%)	p<0·0001
Fungi	**266/1,960 (13·57%)**	**5,839/13,314 (43·86%)**	**p<0·0001**
Aspergillus	29/236 (12·29%)	75/1,453 (5·16%)	p<0·0001
Candida	195/364 (53·57%)	5,584/6,655 (83·91%)	p<0·0001
Cryptococcus	29/1,325 (2·19%)	74/3,663 (2·02%)	NS
Pneumocystis	13/35 (37·14%)	106/1,543 (6·87%)	p<0·0001

Data are positive tests/total tests (%). Northern regions included Beijing, Heilongjiang, Shaanxi, Xinjiang, Henan, Liaoning, and Shanxi; southern regions included Hubei, Yunnan, Sichuan, Shanghai, Guangdong, Jiangsu, Zhejiang, Hunan, and Jiangxi. Denominators vary by pathogen and region because testing was clinically directed. CMV=cytomegalovirus; SARS-CoV-2=severe acute respiratory syndrome coronavirus 2; EBV=Epstein–Barr virus; HAdV=human adenovirus; HBoV=human bocavirus; HCoV=seasonal human coronavirus; HMPV=human metapneumovirus; HPIV=human parainfluenza virus; HRV=human rhinovirus; HSV=herpes simplex virus; IFV=influenza virus; RSV=respiratory syncytial virus; MP=Mycoplasma pneumoniae; CP=Chlamydia pneumoniae; MTB=Mycobacterium tuberculosis; NS=not significant.

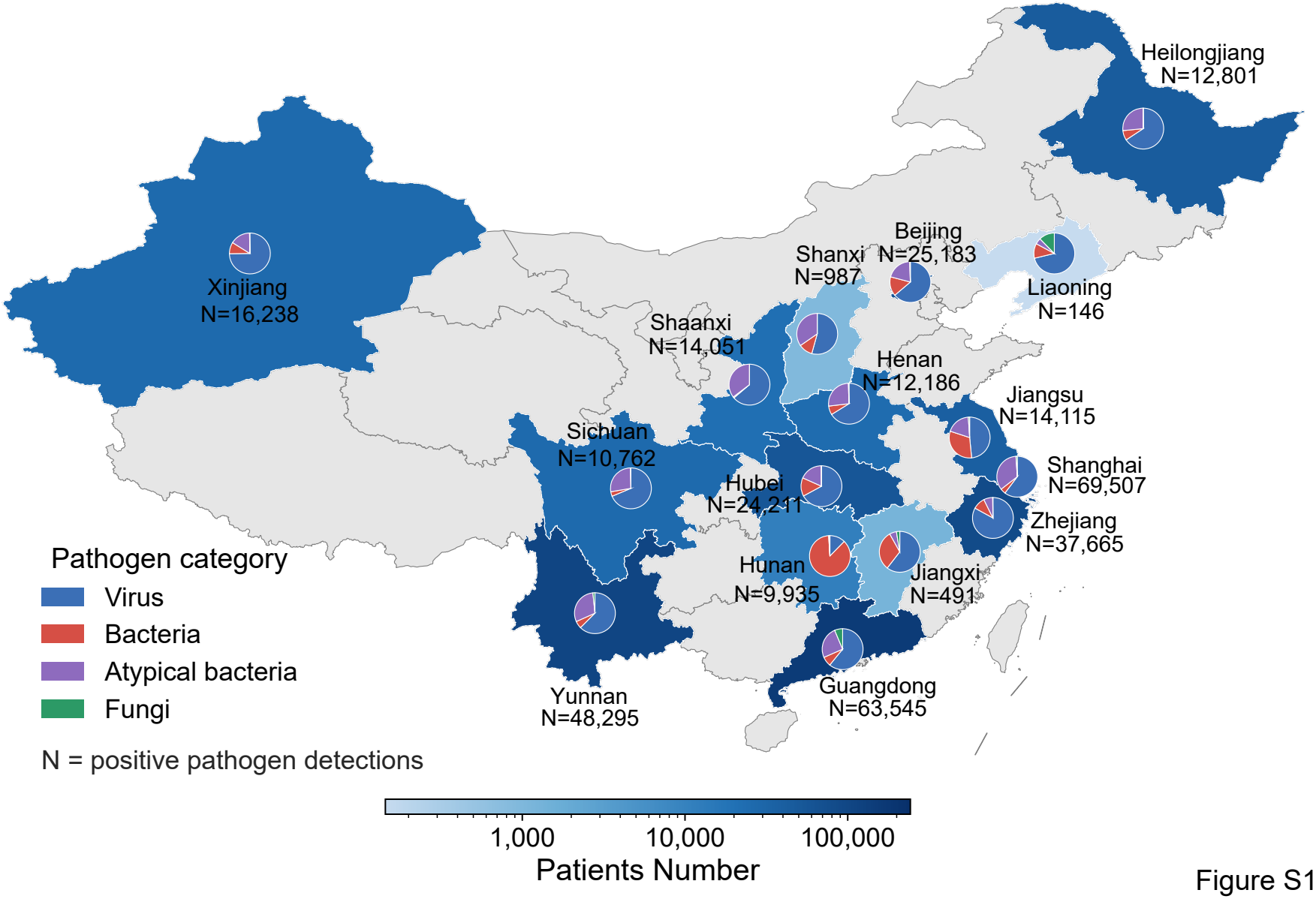


Figure S1

Figure S1: Geographical distribution of included patients and positive pathogen detections across China

Map shading indicates the number of included patients in each provincial-level administrative region. Pie charts show the composition of positive pathogen detections by pathogen category: viruses, bacteria, atypical bacteria, and fungi. N indicates the total number of positive pathogen detections in each region. Grey regions were not represented in the surveillance network.

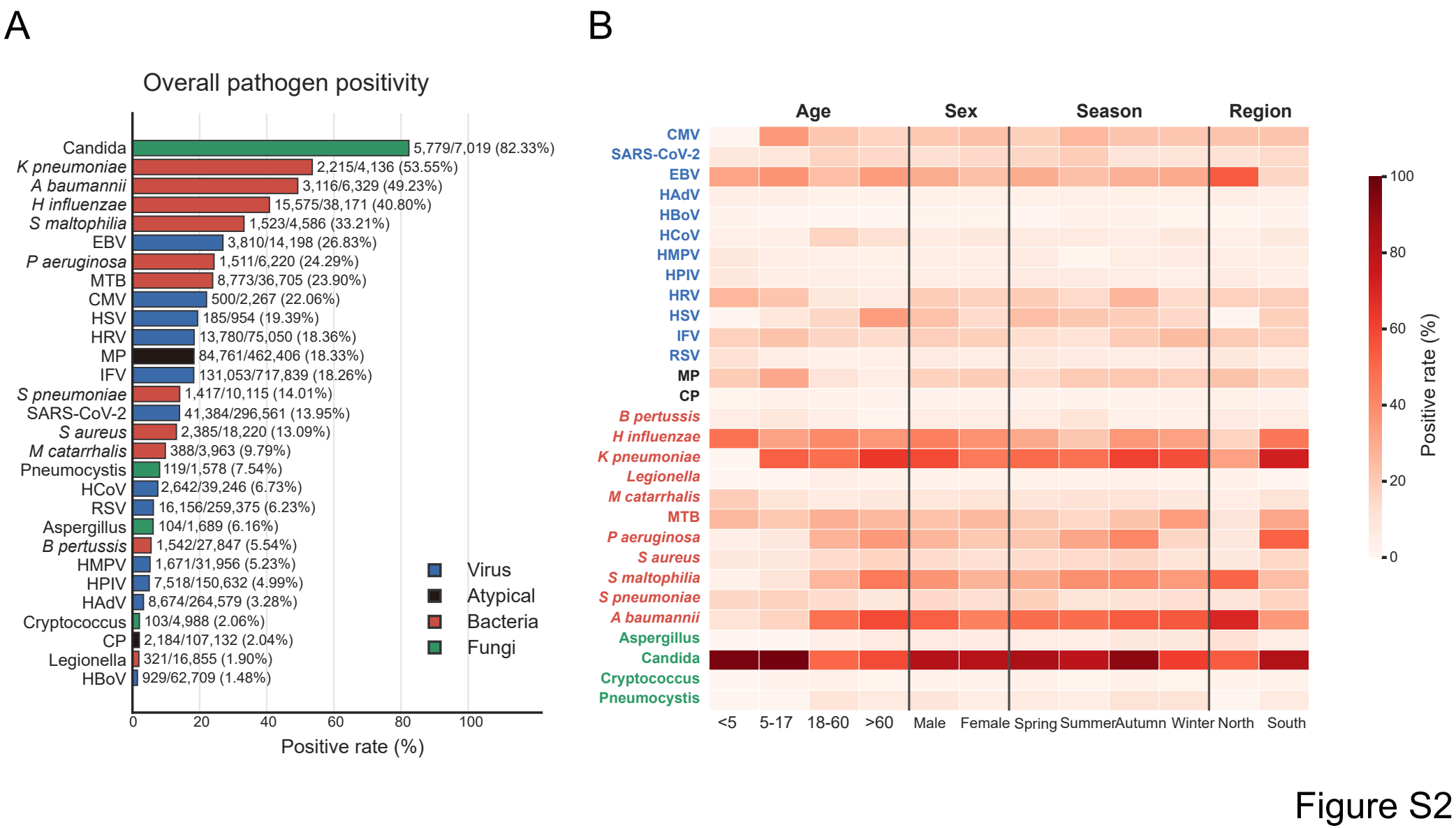


Figure S2

Figure S2: Overall and stratified positivity across 29 respiratory pathogens

(A) Bars show the number positive/number tested and positivity percentage for each pathogen, ordered by positivity. Colors indicate pathogen category. Because testing was clinically directed, positivity should not be interpreted as population prevalence. (B) Heatmap shows pathogen positivity stratified by age group, sex, season, and geographic region. Color intensity represents the positivity percentage within each stratum, with darker red indicating higher positivity. Denominators vary across pathogens and strata because testing was clinically directed.

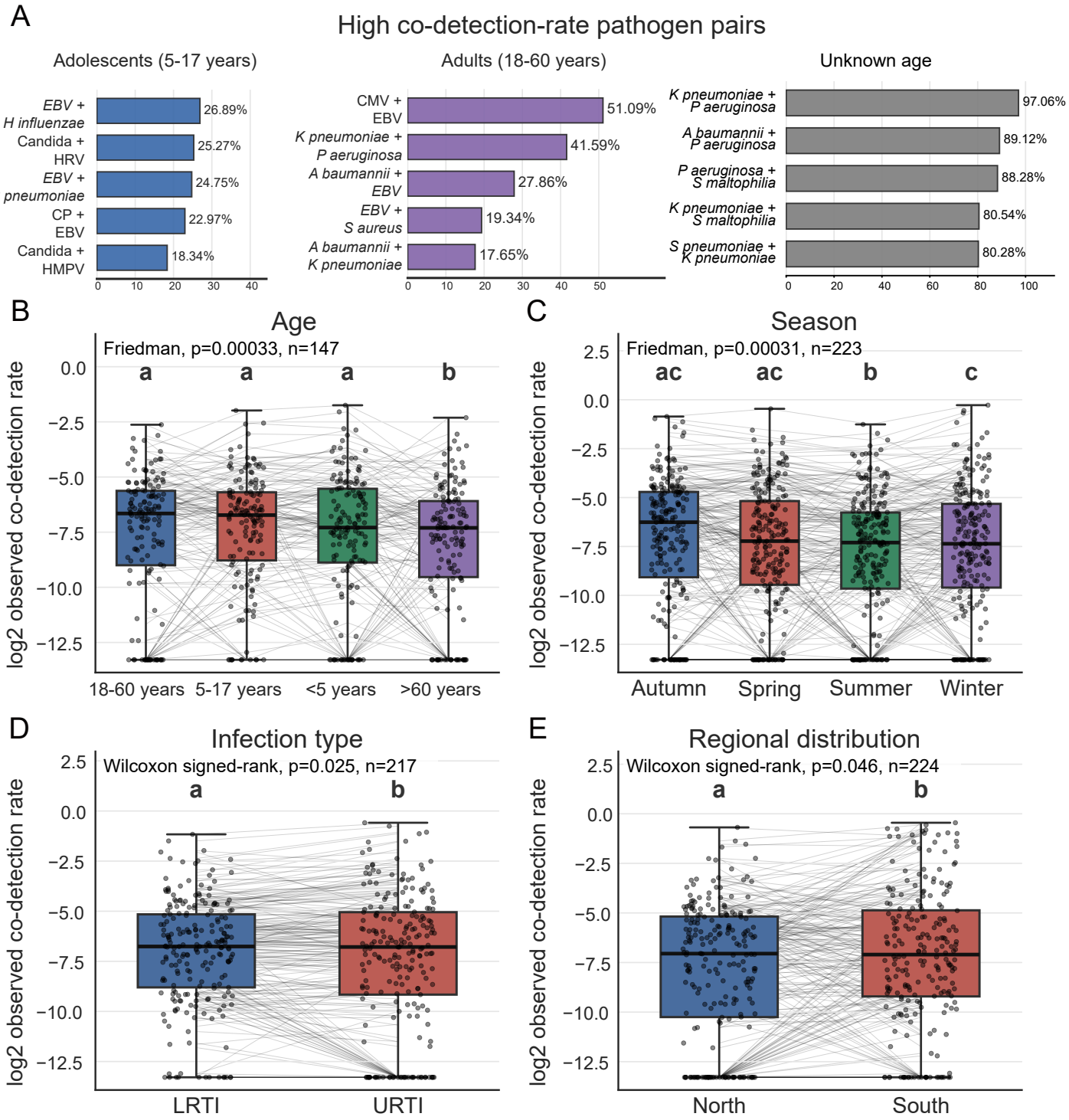


Figure S3

Figure S3: Age-specific leading pairs and stratified observed co-detection

(A) Five pairs with the highest observed co-detection rates in those aged 5–17 years, adults aged 18-60 years and unknown age. Only pathogen pairs with more than 100 jointly tested windows within the corresponding age group were considered for ranking. (B-E) Distributions of log2-transformed observed co-detection rates by age (B), season (C), infection type (D), and region (E). Points represent pathogen pairs, grey lines connect the same pair across strata, and boxplots show medians and interquartile ranges. Panel annotations report global or paired tests and the number of matched pairs. Different letters indicate significant pairwise differences; groups sharing a letter are not significantly different.

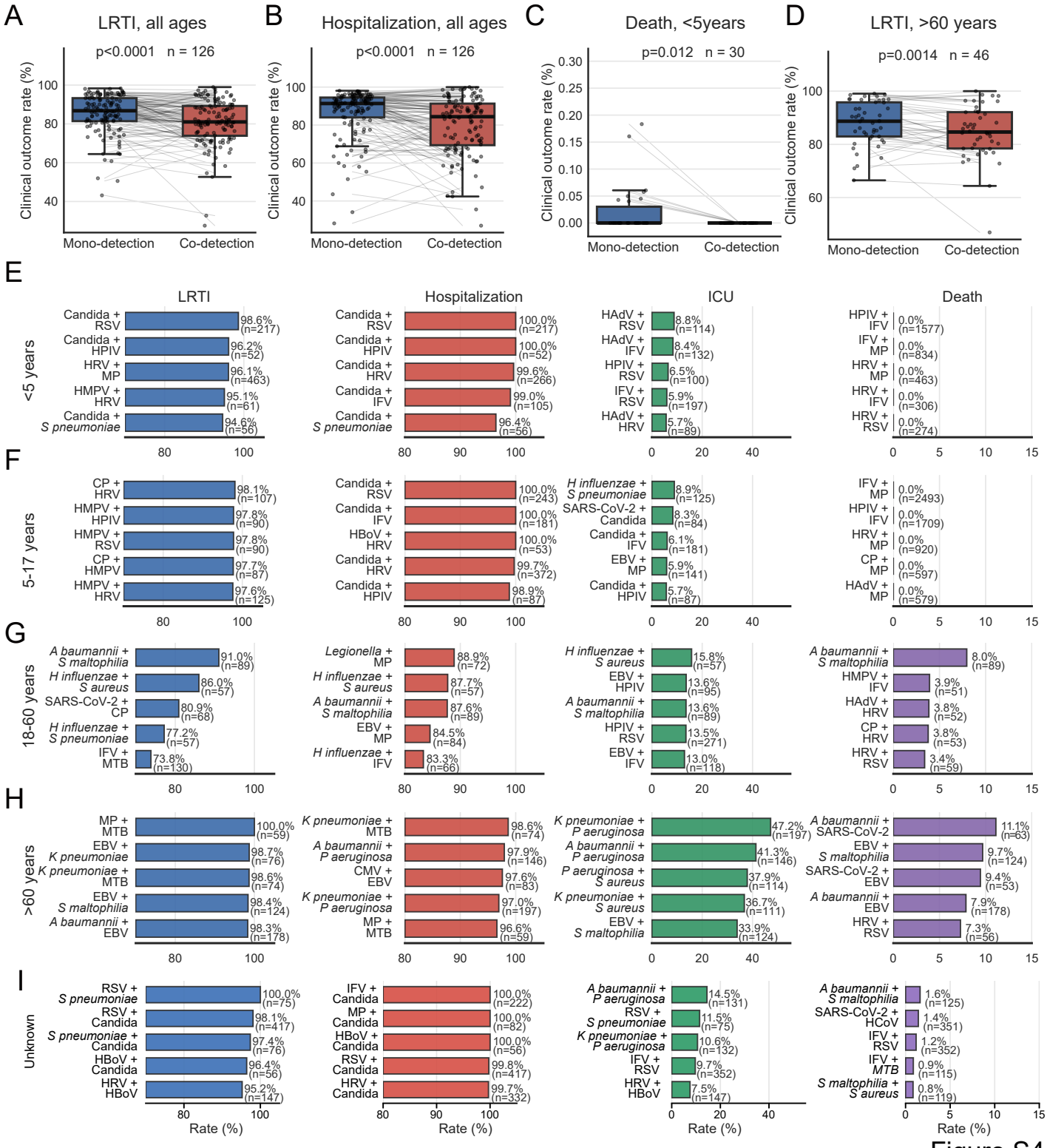


Figure S4: Pair-specific clinical outcome rates across age groups

(A-D) Selected paired comparisons of outcome rates during pair-specific mono-detection and co-detection: all-age LRTI (A), all-age hospital admission (B), death in children younger than 5 years (C), and LRTI in adults older than 60 years (D). Reference groups are defined as in figure 2. Points represent eligible pairs, grey lines connect the two rates for each pair, and boxplots show medians and interquartile ranges. P values are from two-sided Wilcoxon signed-rank tests. (E-I) Five eligible pairs with the highest co-detection-associated LRTI, hospital-admission, ICU-admission, and death rates in children younger than 5 years (E), those aged 5–17 years (F), adults aged 18-60 years (G), adults older than 60 years (H), and unknown age (I). Rates use outcome-known co-detection windows; numbers in parentheses indicate the total number of co-detection windows for each pathogen pair, whereas outcome rates were calculated among windows with known status for the corresponding outcome.

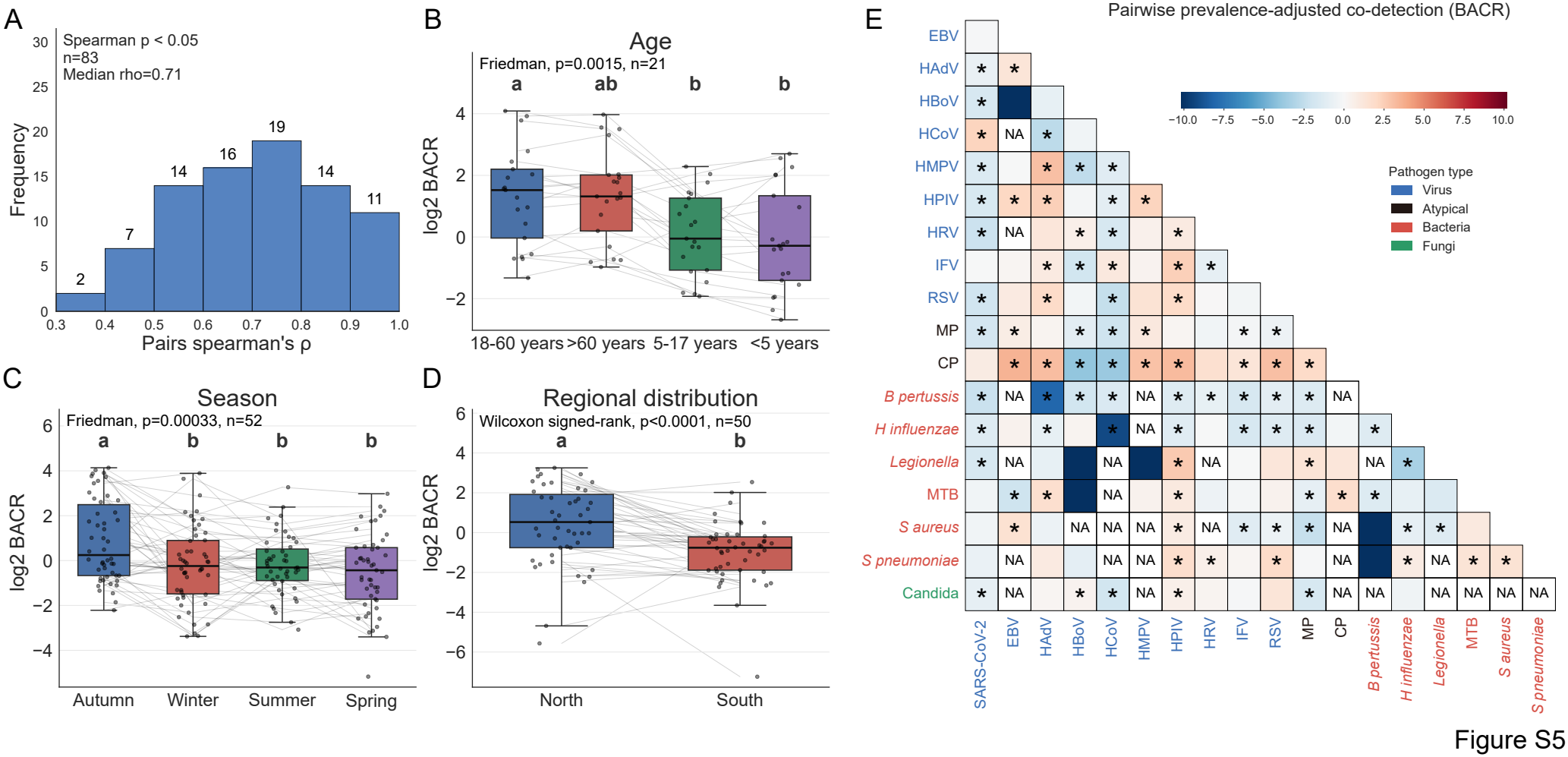


Figure S5

Figure S5: Marginal-positivity dependence, stratified BACR, and hospital-panel balancing

(A) Spearman correlations between observed and expected monthly co-detection among pairs with nominal P<0·05 and ρ>0·3 (83 pairs; median ρ=0·71). (B-D) Distributions of log2(BACR) by age (B), season (C), and region (D). Points represent pairs, grey lines connect the same pair, and boxplots show medians and interquartile ranges. Annotations report Friedman or Wilcoxon signed-rank tests and numbers of matched pairs. Different letters indicate significant pairwise differences. (E) BACR in the hospital-panel-balanced cohort. Colours show log2(BACR); asterisks indicate Benjamini-Hochberg FDR q<0·05 and NA insufficient support. Label colours indicate pathogen category.

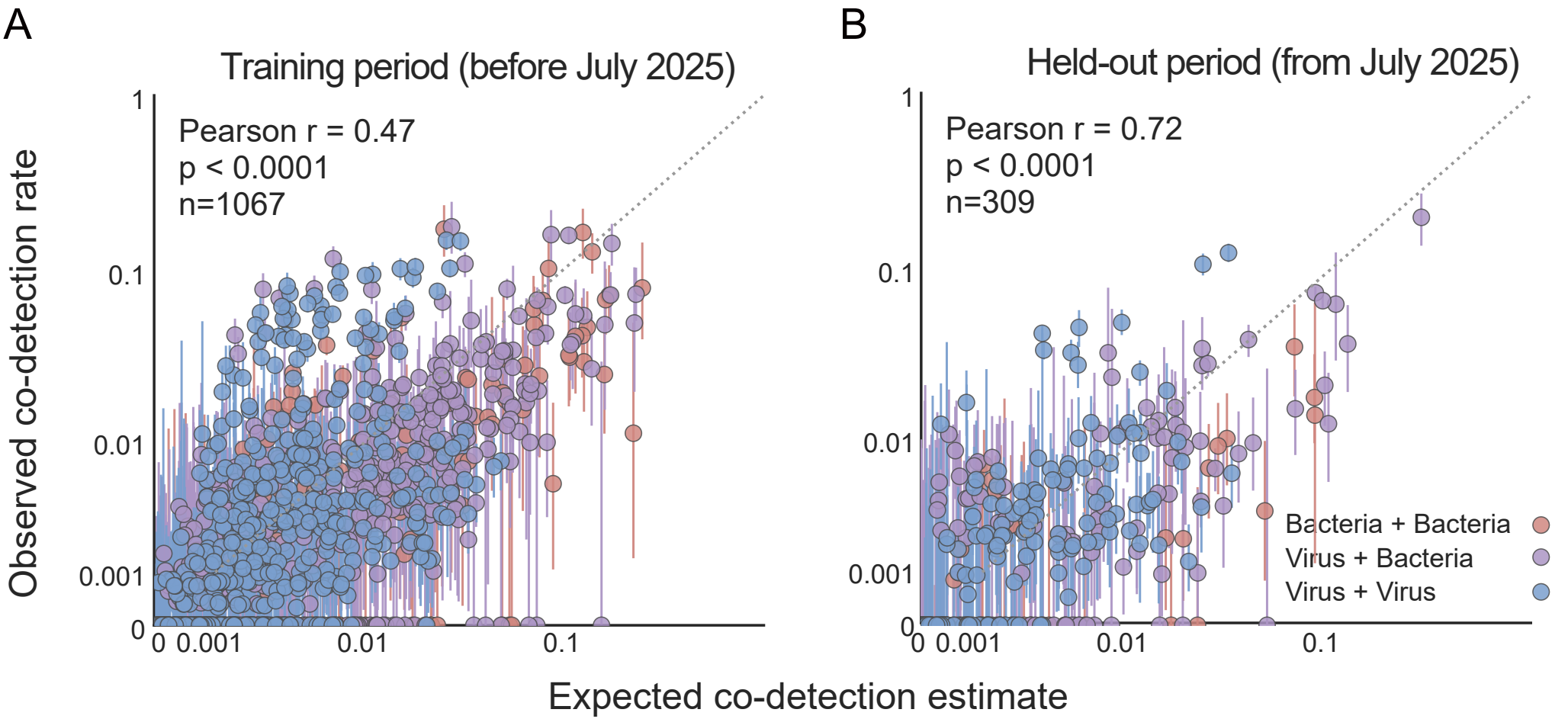


Figure S6

Figure S6: Temporal performance of expected co-detection rates under statistical independence

(A–B) Agreement between observed monthly co-detection rates and co-detection rates expected under statistical independence during the parameterization period before July 2025 (A) and the held-out temporal test period from July 2025 onwards (B). Expected co-detection was calculated from contemporaneous marginal positivity as P_t(a)×P_t(b), without BACR adjustment. Each point represents an evaluable pathogen pair-month among pairs with an FDR-significant training-period BACR; colours indicate bacteria–bacteria, virus–bacteria, and virus–virus pairs. Vertical lines show 95% beta intervals for observed co-detection, and dotted diagonal lines indicate perfect agreement. Panel annotations report Pearson correlation coefficients between expected and observed co-detection rates and the number of pair-month observations.

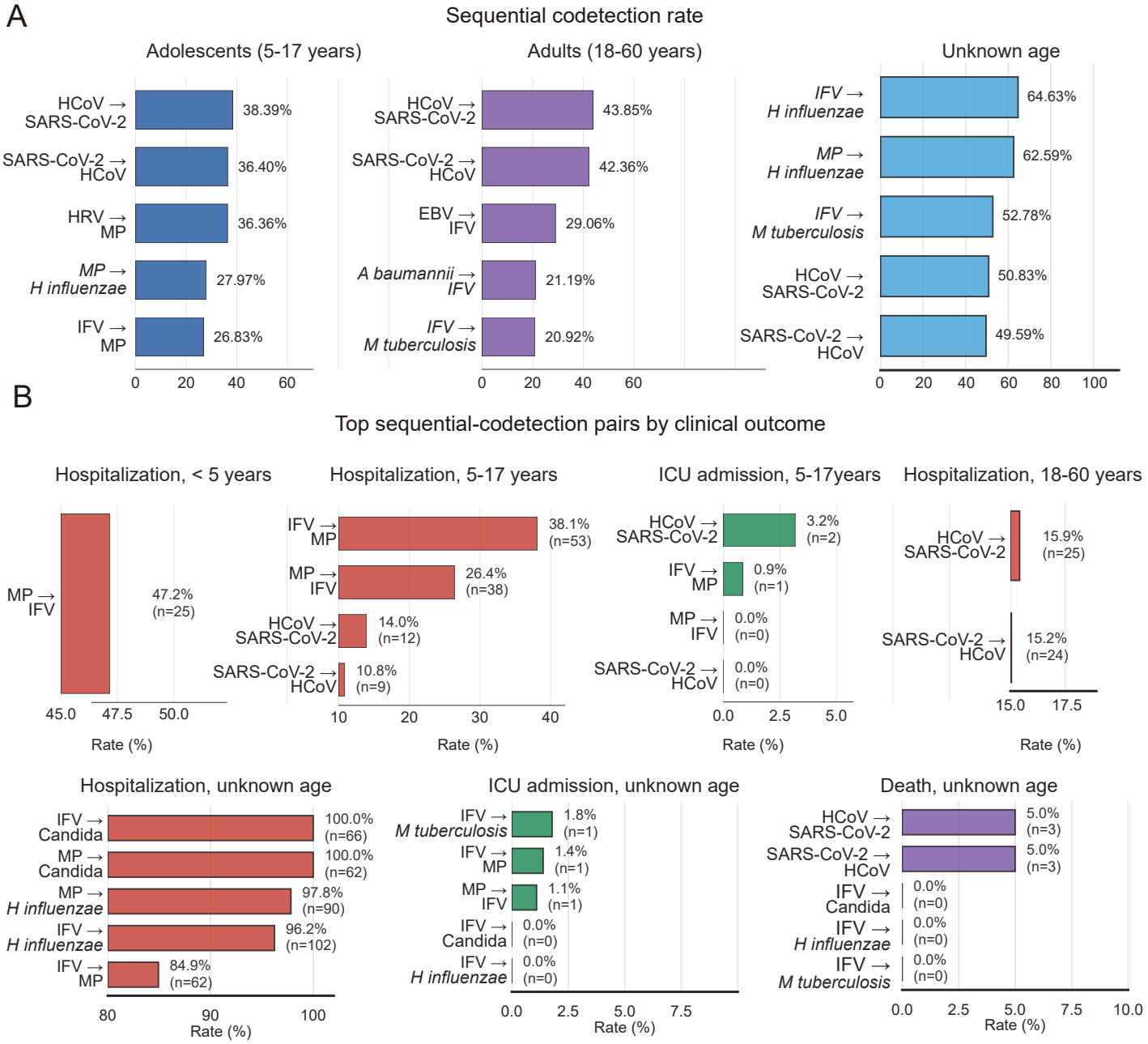


Figure S7

Figure S7: Age-specific sequential pathogen detection rates and clinical outcome patterns

(A) Pathogen pairs with the highest sequential detection rates among adolescents aged 5–17 years, adults aged 18-60 years and patients with unknown age. Bars show the proportion of subsequent windows positive for pathogen B among transitions in which pathogen A was positive in the preceding window and pathogen B was tested in the subsequent window. Only ordered pairs with at least 100 eligible transitions were considered for ranking. (B) Age-specific pathogen pairs with the highest clinical outcome rates following sequential detection. Bars show hospitalisation, ICU-admission or death rates among B-positive subsequent windows for eligible ordered pathogen pairs. Only comparisons meeting the prespecified minimum sample-size criterion were included. Numbers in parentheses indicate the number of outcome events among B-positive subsequent windows (the numerator of the corresponding outcome rate)

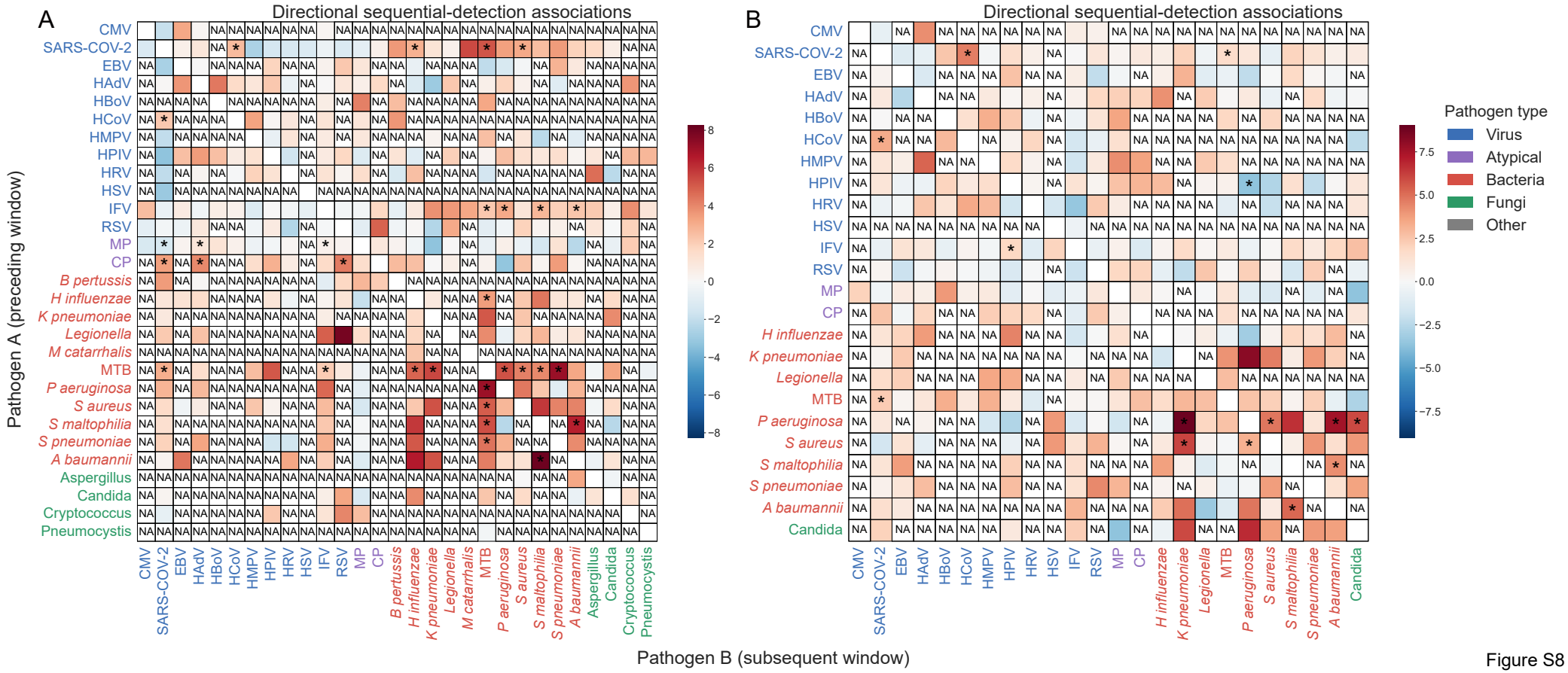


Figure S8

Figure S8: Sequential pathogen detection associations in non-inpatient and preceding-ICU transitions

(A) Sequential-detection associations in non-inpatient transitions. (B) Sequential-detection associations in transitions whose preceding window was an inpatient ICU window.

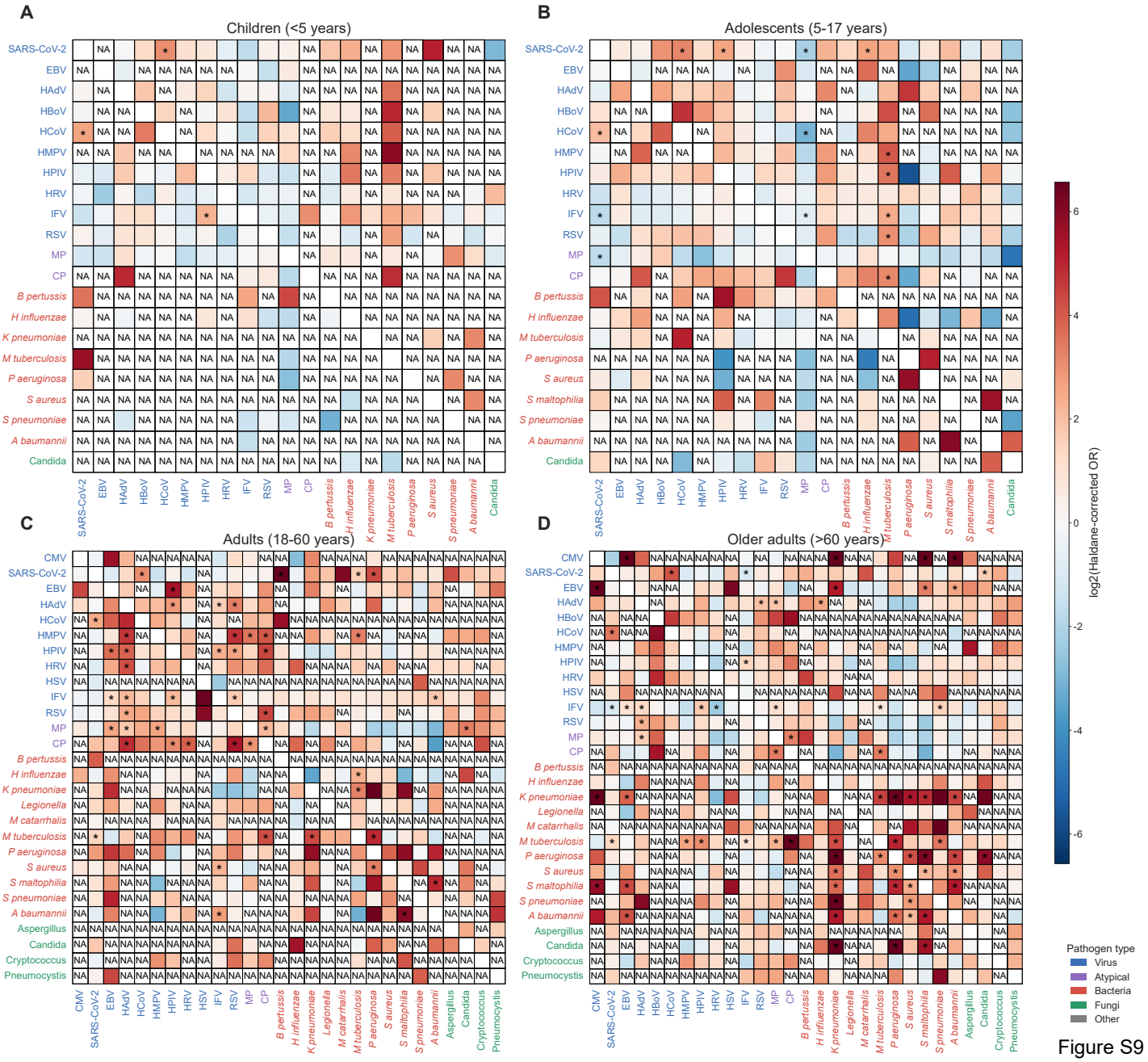


Figure S9

Figure S9: Age-stratified sequential pathogen detection associations

Cross-pathogen sequential-detection ORs for children younger than 5 years (A), adolescents aged 5–17 years (B), adults aged 18–60 years (C), and adults older than 60 years (D). Rows denote the source pathogen in the preceding window and columns the target pathogen in the subsequent window. For each ordered pair A(t)→B(t+1), the OR compares subsequent B positivity after A-positive versus A-negative preceding windows, restricted to transitions in which A was tested at t and B was tested at t+1. Baseline B negativity was not required. Cell colours show log2-transformed Haldane–Anscombe-corrected ORs; red indicates OR>1 and blue OR<1. Asterisks indicate Benjamini–Hochberg FDR q<0·05, and NA indicates a non-estimable comparison. Pathogen-label colours indicate pathogen category.